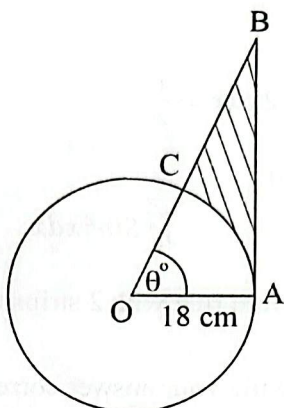


Section A [40 Marks]

Answer all questions.

- 1 (a) Simplify $\frac{x^3\sqrt{x}}{3x^5 \times x^{-2}}$ [2]
- (b) Y is inversely proportional to the cube root of $x + 1$.
Given that $y = 2$ when $x = 63$, express y in terms of x . [2]
- 2 A sequence $U_1, U_2, U_3, \dots$ is given by $U_{n+1} = 3 - U_n$.
- (a) Describe the behavior of the sequence when $U_1 = 2$. [2]
- (b) Show that $U_{n+2} = U_n$ and find an expression for U_{n+3} in terms of U_n . [2]
- 3 Paul is given an interest-free loan to buy a second hand car. He pays the loan in monthly instalments. He pays \$20 for the first month, \$22 the second month and the payments continue to rise by \$2 per month until the loan is repaid.
Given that the final monthly repayment is \$114,
- (a) find the number of months it will take to repay the loan. [2]
- (b) Hence find the amount of the loan. [2]
- 4 The functions f and g are defined by
- $$f: x \longrightarrow 2 - \frac{1}{x}$$
- $$g: x \longrightarrow \frac{1}{2x+6}.$$
- (a) Find,
- (i) gf , [2]
- (ii) $f^{-1}(x)$. [2]
- (b) Describe the geometrical relationship between the graphs of $g(x)$ and $g^{-1}(x)$. [1]

- 5 The diagram shows a circle with centre O. The radius of the circle is 18 cm and angle AOC is θ .



- Find the area of the shaded region ACB, giving your answer in terms of θ . [5]
- 6 Express $\frac{x^2+3x-13}{(x+2)(x-1)}$ in the form $A + \frac{B}{x+2} + \frac{C}{x-1}$ where A, B and C are constants. [5]
- 7 The polynomial $P(x) = x^3 - ax^2 + x - b$ is exactly divisible by $x - 1$ and leaves a remainder of -1 when divided by $x - 2$.
- (a) Find the values of a and b . [4]
- (b) Factorise $P(x)$ completely. [2]
- 8 Mr Moyo stays at point A with coordinates $(-5;3)$. A well P with coordinates $(2;1)$ is at the midpoint of a straight road joining Mr Moyo's house to Mr Dube's house.
- Find
- (a) the coordinates of Mr Dube's house, [2]
- (b) the distance from Mr Moyo to Mr Dube's house. [2]
- (c) The equation of a line of points which are of equal distance from Mr Moyo and Mr Dube's houses. [3]



Section B [80 Marks]

Answer any five questions.

9 (a) Show that

$$\int_0^{\frac{\pi}{4}} \cos 2x dx = \frac{1}{2}. \quad [2]$$

(b) Find the exact value of

$$\int_0^{\frac{\pi}{4}} \sin^2 x dx. \quad [4]$$

(c) (i) Use the trapezium rule with 2 strips to estimate the value of

$$\int_0^{\frac{\pi}{4}} \sec x dx, \text{ giving your answer correct to 2 significant figures.} \quad [4]$$

(ii) Determine by sketching the graph of $y = \sec x$, whether the Trapezium Rule under estimates or over-estimates area under the graph. [3](d) Starting with $x_0 = 2.5$, use the iterative formula,

$$x_{r+1} = \frac{5}{x_r} + 1, \text{ to find } x_1, x_2 \text{ and } x_3. \quad [3]$$

10 (a) It is given $M = \begin{pmatrix} 2 & 1 & -1 \\ -1 & 0 & 3 \\ 2 & K & 4 \end{pmatrix}.$

(i) State the order of matrix M . [1](ii) Find M^2 in terms of K . [2](iii) Find K given that $|M| = 20$. [3]

(b) It is given that matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}, A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & -2 & 1 \end{pmatrix} \text{ and}$$

$$B^{-1} = \begin{pmatrix} 1 & -4 & 10 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

(i) Find $(AB)^{-1}$. [3]

- (ii) Hence or otherwise, solve the following system of equations

$$\begin{array}{rcl} x & = & 1 \\ -x + y & = & 2 \\ 3x + 2y + z & = & 1 \end{array} \quad [3]$$

- (c) Given also that

$$AP = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}$$

Find matrix P. [4]

- 11 (a) (i) Prove that $\frac{1-2\cos^2\theta}{\sin\theta\cos\theta} \equiv \tan\theta - \cot\theta$. [3]

- (ii) Hence solve the equation

$$\frac{1-2\cos^2\theta}{\sin\theta\cos\theta} = 2,$$

giving all values of θ between 0° and 360° . [5]

- (b) A circle is defined parametrically by the equations

$$\begin{aligned} x &= 3 + 2\cos\theta \\ y &= 4 + 2\sin\theta, \end{aligned}$$

where θ is a real parameter.

- (i) Show that $\frac{dy}{dx} = -\cot\theta$. [2]

- (ii) Show that the Cartesian equation of the curve is given by

$$(x-3)^2 + (y-4)^2 = 4 \quad [4]$$

- (iii) Describe fully the curve given in (ii). [1]

- (iv) Given that the line $y = 2x$ meets the curve in (ii) at A and B, find the coordinates of A and B. [3]



- 12 The position vectors of the villages A, B and C are $-2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$, $-2\mathbf{i} - \mathbf{j} - 3\mathbf{k}$ and $-\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$ respectively.

(a) Find the distance between village A and village B. [3]

(b) During a survey, the engineer surveyed a straight line from village A to village C.

Find the vector equation of the line in parametric form. [4]

(c) The engineer surveyed another straight line from village B to village C. A borehole is to be drilled somewhere in between the surveyed lines.

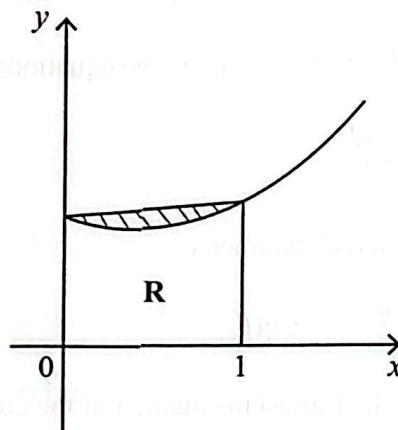
Calculate the maximum angle that makes the position of the borehole to lie anywhere between the surveyed lines. [4]

(d) The three villages are represented in the Cartesian plane.

Find the equation of the plane containing the three villages. [5]

- 13 (a) Find $\int e^{3x} \sin 4x dx$. [6]

(b)



The diagram shows part of the curve $y = 1 + x^2$. The region R is bounded by the curve, the axes and the line $x = 1$,

- (i) Show that the volume of the solid formed when R is rotated completely about the x -axis is $\frac{28\pi}{15}$ units. [4]
- (ii) Use the Trapezium Rule with 4 trapezia to estimate the area of R, giving your answer correct to 3 s.f. [4]
- (iii) State with a reason whether the Trapezium rule method overestimates or underestimates the area. [2]



- 14 In Mt Darwin, an ironsmith uses the traditional smelting method to heat the metal using a wood powered oven. He produces hoes and axes for domestic use. It is assumed that at t minutes, the rate of change of temperature of a cooling metal is proportional to the temperature X °C of that metal at the time it is removed from the oven. Initially $X = 82$ °C.

(a) Show that,

$$X = 82^{-kt}. \quad [6]$$

(b) Given also that, $X = 42$ °C, when $t = 10$ minutes find how much longer it will take the metal to cool to 37 °C under these conditions. [4]

(c) (i) Estimate the temperature of the metal after 3.2 minutes out of the oven. [2]

(ii) Suggest reasons why the ironsmith heats the metal during the process of producing hoes and axes. [1]

(d) Given that

$$\ln y = xy, y > 0, \text{ show that, } \frac{dy}{dx} = \frac{y^2}{1-xy}. \quad [3]$$

- 15 (a) It is given that $\sin x = x^2 e^y$.

$$\text{Show that } \frac{dy}{dx} = \frac{\cos x}{x^2 e^y} - \frac{2}{x}. \quad [2]$$

(b) If $x = \frac{t^2}{1-t}$ and $y = \frac{t}{1+t}$,
find $\frac{dy}{dx}$ in terms of t . [3]

(c) Find by differentiation, the coordinates of the turning points on the curve $y = 4x^3 - x^2 - 4x + 10$, and determine their nature. [6]

(d) Given that

$$f(x) = e^x \sin x,$$

find the first 3 terms in the Maclaurin series for $e^x \sin x$. [5]

